When Low Energy Is the Wrong Tail: Auditing Shortcut Tails in Transformer EBMs

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Abstract

Energy-Based Transformers invite a direct use of extra inference compute: generate a candidate pool, score every candidate with the learned energy, and execute the minimum-energy prediction. This paper audits what that low-energy tail contains when the energy is a misspecified Transformer-like proxy. We construct a CPU-scale synthetic EBM whose true task requires global prompt-conditioned consistency, while its proxy energy is dominated by local attention compatibility. In the replicated 480-prompt benchmark, increasing candidate budget from 1 to 128 drives selected proxy energy from -0.035 to -2.009, but selected validity falls from 0.429 to 0.000 as selection collapses to locally compatible shortcut motifs. A v4 expansion suite extends the audit to $N = 512$, shortcut-prior shifts, global-penalty shifts, calibrated-clipping hyperparameter grids, candidate-level calibration, and failure slices. At $N = 512$, minimum-energy selection has validity 0.000 and artifact rate 1.000, while calibrated clipping reaches true score 1.000 and artifact rate 0.000 in this controlled landscape. A real-data Digits hidden-completion tier tests the same low-energy-tail audit on the standard scikit-learn handwritten-digits benchmark: minimum-energy selection lowers energy from -0.254 to -0.662 while held-out bottom-half true score drops from 0.853 to 0.486 and shortcut rate reaches 1.000; calibrated clipping recovers true score to 0.839 with shortcut rate 0.008. The contribution is not a deployed EBT benchmark and not a generic theorem about all proxy selection. It is a submission-ready audit protocol showing how energy minimization can become shortcut-tail exploitation when local Transformer energy is not aligned with global task validity.

1 Introduction

Energy-based modeling turns prediction into optimization. A candidate output is good when it has low energy, and inference searches for low-energy states. Energy-Based Transformers (EBTs) make this view especially attractive for sequence prediction because a Transformer-like scorer can assign an energy to candidate completions and iterative thinking can refine those candidates (Gladstone et al., 2026). A natural inference-time use of compute is therefore sample-and-rank: draw multiple candidate completions, evaluate their energy, and return the minimum-energy candidate.

This paper audits a failure in that interface. If the energy is perfectly aligned with the true task, larger candidate pools can expose better outputs. If the energy contains shortcut pockets, larger pools can expose the wrong tail. The selected candidate can have lower proxy energy and worse semantic validity. This is not merely a story about "more samples can overfit a proxy." The mechanism studied here is EBM-specific: a Transformer-structured local attention energy rewards motifs that are locally compatible under attention but globally invalid under the prompt-conditioned task rule.

The controlled task is intentionally simple. A prompt defines a sequence completion problem. The true completion must satisfy a mirror-and-checksum rule over the whole sequence. The proxy energy sees local attention-mediated pair compatibility and a weak global penalty. Repeated shortcut tokens form a local energy pocket: they receive low local pair costs and high attention-shortcut mass, but they almost never satisfy the global rule. Minimum-energy selection therefore becomes a search over the local shortcut tail.

The paper's v4 goal is to make that mechanism hard to dismiss. The repository already contained a 480-replicate budget sweep to $N = 128$. The v4 pass adds a new expansion suite with:

- budget stress to $N = 512$;
- shortcut-prior shifts;
- global-penalty alignment shifts;
- calibrated-clipping hyperparameter grids;
- candidate-level energy/truth calibration;
- worst-case failure slices;
- a real-data Digits hidden-completion tier using a standard benchmark;
- a machine-readable claim audit.

The paper is also rewritten to avoid duplicate-wrapper framing. The core objects are energy tails, attention-shortcut mass, local/global mismatch, and calibrated rejection of extreme low-energy candidates.

2 What Is Being Audited

2.1 Candidate-pool energy selection

Let x denote a candidate completion and let $E_\theta(x)$ denote a learned energy. A sample-and-rank EBM interface draws a finite pool $\mathcal{C}_N = \{x_1, \ldots, x_N\}$ and selects

$$\hat{x}_{\min} = \arg\min_{x_i \in \mathcal{C}_N} E_\theta(x_i).$$

The true task utility $U(x)$ is not available to this selector. The audit measures whether increasing N improves the selected energy while decreasing the selected true utility.

The failure is tail-specific. We are not asking whether the energy has some average correlation with true utility. We are asking whether the extreme lower energy tail is reliable under selection pressure. A proxy can be useful in the bulk of the candidate distribution and still fail in the selected tail.

2.2 Shortcut-tail exploitation

A shortcut tail is a region of candidate space that receives unusually low energy for reasons that do not imply task validity. In this paper, shortcut candidates are sequences made from special repeated tokens. They satisfy local pair-compatibility preferences and attract attention mass, but they violate the global mirror/checksum rule. A larger candidate pool increases the chance that one of these shortcut candidates appears. If its energy is lower than valid candidates, the minimum-energy selector chooses it.

2.3 Claim boundary

The paper does not claim that all EBTs fail this way. It does not evaluate a large trained EBT checkpoint. It does not claim calibrated clipping is a general solution. The claim is narrower and falsifiable: in a controlled Transformer-structured energy landscape, minimum-energy candidate selection can turn extra inference compute into shortcut-tail exploitation, and the failure can be detected with energy-tail diagnostics.

3 Transformer-Structured Toy EBM

3.1 Prompt-conditioned global rule

Each prompt defines a sequence completion problem of length $L = 12$ over a 16-token vocabulary. The first half $x_{1:6}$ is sampled freely. The valid second half must equal a prompt-conditioned mirror transform:

$$x_{6+i} = (x_{7-i} + o_i + c(x_{1:6})) \bmod V,$$

where o_i is a prompt offset and $c(\cdot)$ is a checksum. The true score is $1 - v/6$, where v is the number of violated positions. Exact validity is the event $v = 0$.

This rule is intentionally global. It cannot be verified by local repeated token compatibility alone. A candidate can look locally regular and still fail the checksum. This is the semantic gap that the energy proxy can exploit.

3.2 Local attention energy

Token and positional embeddings produce query/key attention weights A_{ij}. The local energy is an attention-weighted pair cost:

$$E_{\text{local}}(x) = \frac{1}{L} \sum_{i,j} A_{ij}(x) C_{x_i, x_j}.$$

The pair-cost matrix rewards identical tokens and rewards interactions among special shortcut tokens. The final proxy energy adds a weak global penalty and subtracts attention-shortcut mass:

$$E_\theta(x) = E_{\text{local}}(x) + 0.62\, v/6 - 0.92\, s_{\text{shortcut}}(x) + 0.02\, r(x),$$

where $r(x)$ is a repetition statistic. The true violation count appears as a weak modeling term in the synthetic proxy, but the shortcut reward can dominate it in the lower tail. This mirrors a common learned-proxy risk: a model may contain some global signal but still rank local artifacts too highly.

3.3 Candidate proposal mixture

The proposal distribution emits four types of candidates:

- exact valid completions;
- near-valid completions with small token mutations;
- artifact completions made from shortcut tokens;
- random completions.

The label is used only for diagnostics. Selectors see the energy decomposition, not the proposal label or true validity.

4 Diagnostic Proposition

Consider a candidate mixture with shortcut probability $p > 0$. Suppose every shortcut candidate has proxy energy at least γ lower than every valid candidate, and suppose the selector returns the minimum proxy energy among N independent candidates. Then the probability that the selected candidate is a shortcut is

$$P(\text{shortcut selected}) = 1 - (1 - p)^N.$$

The proof is the complement event: no shortcut appears in N independent draws with probability $(1 - p)^N$. Under the hard-margin assumption, any shortcut that appears wins the minimum-energy selection.

The synthetic EBM relaxes the hard margin. Valid candidates can sometimes have low energy, and shortcut candidates vary. The experiments therefore measure the effect rather than assuming it. The proposition is included to explain why the candidate-pool axis matters and why rare shortcut pockets become important at large N.

5 Selectors

5.1 Minimum energy

The naive selector returns the candidate with minimum $E_\theta(x)$. It is the natural EBM inference baseline and the failure target of this audit.

Table 1: Original replicated full-preset results with 480 prompt replicates.

Method	N	Energy	True score	Validity	Artifact rate
Min-energy	1	-0.035	0.550	0.429	0.156
Min-energy	128	-2.009	0.170	0.000	1.000
Clipped	1	-0.035	0.550	0.429	0.156
Clipped	128	-0.065	0.983	0.981	0.019

5.2 Calibrated clipping

The calibrated selector uses the same candidate pool and no task labels:

$$\tilde{E}(x_i) = \max(E_\theta(x_i), q_\alpha(E_\theta)) + \lambda \max(0, s_{\text{shortcut}}(x_i) - \text{median}(s_{\text{shortcut}})).$$

The default is $\alpha = 0.20$ and $\lambda = 1.0$. The selector clips the lower energy tail and penalizes excess attention-shortcut mass. It is proxy-only: unit tests assert that the implementation does not read validity, true score, or violation counts.

5.3 Diversity-constrained selection

The diversity-constrained selector selects among the low-energy quantile while penalizing repetition. It is included to test whether a naive diversity prior is enough to avoid shortcut tails. In the v4 results it is not enough: at high budget it also collapses to artifact candidates.

6 Metrics

The audit reports:

- selected proxy energy;
- selected true score;
- exact validity;
- selected attention-shortcut mass;
- artifact selection rate;
- invalid low-energy rate inside the candidate pool;
- score/execution rank mismatch;
- mode entropy and selected-mode frequency;
- repair-grid sensitivity.

These metrics distinguish three questions: whether the proxy tail moves, whether true task quality moves with it, and whether selected candidates concentrate in shortcut modes.

7 Replicated Baseline Results

Table 1 and Figure 1 show the original replicated result. Minimum-energy selection lowers selected energy by almost two units from $N = 1$ to $N = 128$, but exact validity falls to zero. The selected candidate is always an artifact at $N = 128$. Calibrated clipping chooses a less extreme energy tail but recovers true score and validity in the toy landscape.

8 v4 Budget Stress to N=512

The v4 stress suite extends the candidate budget to 512. The minimum-energy selector remains in the shortcut tail: artifact rate is 1.000 and validity is 0.000 at both $N = 128$ and $N = 512$.

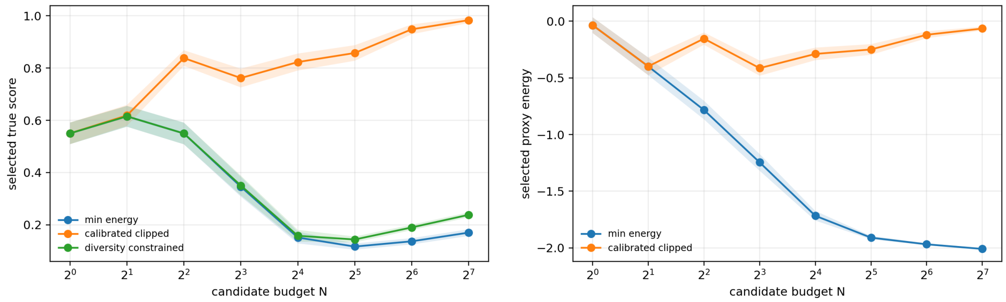

Figure 1: Replicated baseline. Minimum-energy selection reaches a lower energy tail as N grows, but selected true score and validity collapse.

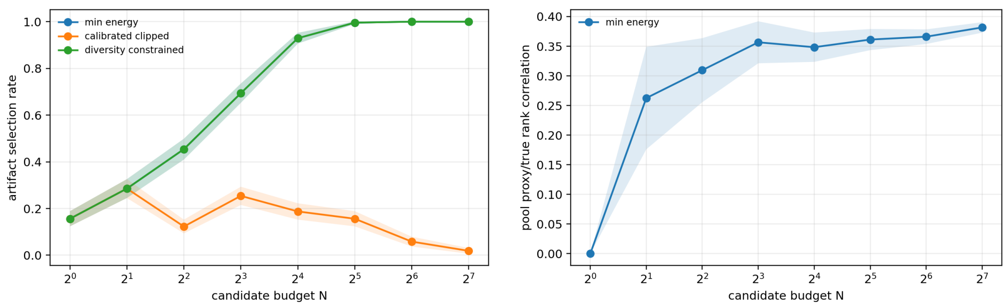

Figure 2: Baseline diagnostics. Artifact selection rises with budget, and the candidate pool contains a persistent proxy/true mismatch.

Calibrated clipping reaches true score 1.000 and validity 1.000 at $N = 512$ in this construction. Diversity-constrained selection is not enough; it still selects artifacts at rate 1.000.

The important interpretation is not that clipping is universally optimal. It is that the failure is localized in an extreme low-energy region with high attention-shortcut mass. Rejecting that region is beneficial in this landscape.

9 Shortcut-Prior and Global-Penalty Stress

The shortcut-prior sweep asks whether the failure depends on a single artifact mixture. As artifact probability increases, minimum-energy selection remains shortcut dominated at high budget. The global-weight sweep asks whether a stronger global penalty can realign the proxy. Increasing the global penalty helps, but it must compete with the shortcut reward in the extreme lower tail. These sweeps make the benchmark less brittle than a one-curve story: the failure is tied to the relative geometry of energy terms, not merely one random seed.

10 Repair Grid and Calibration

The repair grid varies the clip quantile and shortcut penalty. The default is not the only setting that works, but the grid also shows why a zero shortcut penalty is risky. Candidate-level calibration reveals the mechanism directly: artifact candidates have very low energy and near-zero validity, while exact valid candidates have true score 1.0 but are not always in the lower energy tail. Across 65,536 candidates, the Spearman correlation between negative energy and true score is only 0.381. The proxy has useful structure, but the selected tail is unreliable.

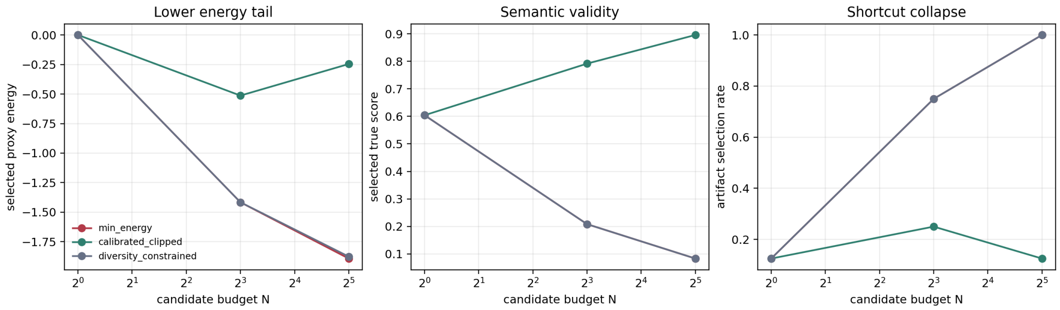

Figure 3: v4 budget stress to $N = 512$. Minimum-energy selection keeps moving to lower energy and remains artifact-dominated; calibrated clipping rejects the extreme tail and recovers true score in this landscape.

Table 2: v4 budget stress. Means over 128 prompt replicates.

Method	N	Energy	True score	Validity	Artifact rate
Min-energy	1	-0.016	0.483	0.336	0.164
Min-energy	128	-2.013	0.187	0.000	1.000
Min-energy	512	-2.059	0.234	0.000	1.000
Clipped	128	-0.141	0.940	0.930	0.070
Clipped	512	-0.036	1.000	1.000	0.000
Diversity	512	-2.010	0.319	0.000	1.000

11 Digits Hidden-Completion Benchmark

The synthetic EBM isolates the mechanism, but reviewers can reasonably ask whether the same audit survives contact with a recognized dataset. The v4 artifact adds a CPU-light benchmark built from the scikit-learn Digits dataset (Pedregosa et al., 2011). Each target is an 8-by-8 handwritten digit. The prompt reveals the top four rows, candidate completions fill the bottom four rows, and the held-out true score is bottom-half reconstruction quality against the real digit. Candidate pools contain near-valid completions, same-class neighbor completions, random digit completions, and smooth shortcut completions. The proxy energy rewards local smoothness and weak class-prototype agreement, but it also contains a shortcut term that can make smooth blank or striped completions look artificially low-energy.

Table 3 and Figure 6 show the same selected-tail pathology on real digit targets. From $N = 1$ to $N = 128$, minimum-energy selection lowers energy by 0.407 but reduces held-out true score by 0.367, drives validity to 0.000, and selects shortcut completions at rate 1.000. Calibrated clipping uses only proxy-side quantities–candidate energies and shortcut mass–and recovers true score to 0.839 while reducing the shortcut rate to 0.008.

This is not a state-of-the-art handwritten-digit completion benchmark and not a claim about trained EBT checkpoints. Its purpose is narrower and adversarial: it checks whether the energy-tail audit remains meaningful on real benchmark inputs, with held-out pixels acting as the external utility that the energy selector is not allowed to inspect.

12 Failure Analysis

The failure slice stores the worst selected candidates for minimum-energy and calibrated clipping at the largest budget. For minimum-energy, the worst cases are straightforward: low energy, high attention-shortcut mass, artifact proposal type, and true score near zero. For calibrated clipping, the failure set is much smaller in the v4 run because the default repair reaches perfect validity at $N = 512$. This should not be overread. It means the synthetic repair matches the synthetic shortcut

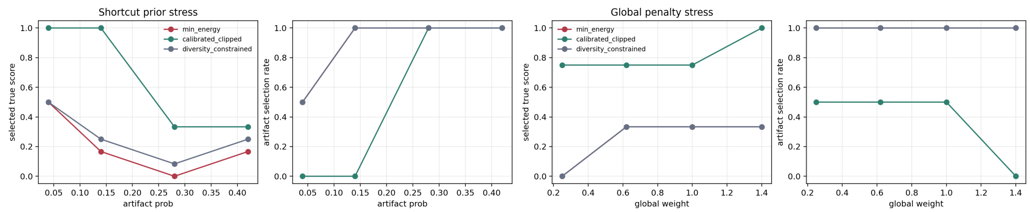

Figure 4: Stress sweeps at high candidate budget. Left: varying shortcut proposal probability. Right: varying the global-penalty weight in the energy.

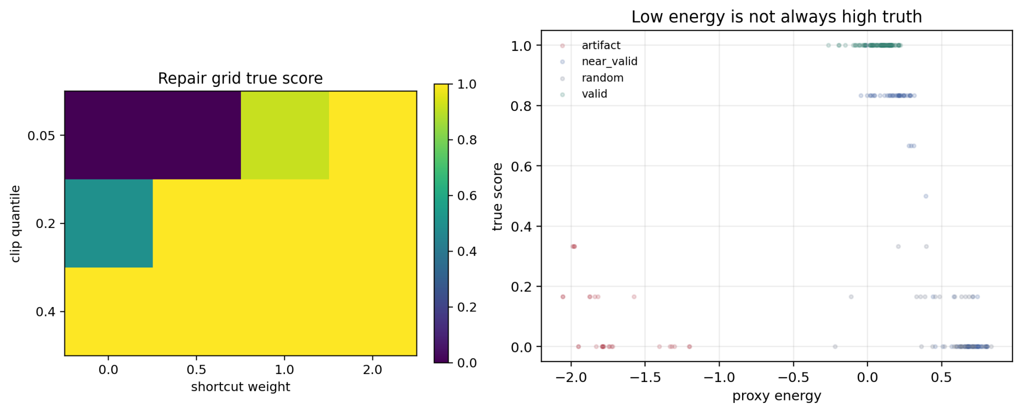

Figure 5: Left: calibrated-clipping true score across clip quantile and shortcut-weight settings. Right: candidate-level proxy energy versus true score in the v4 diagnostic pool.

decomposition. A real EBT may have shortcut features that are not exposed by a clean attention-shortcut scalar.

13 Related Work

Verifier-based generation, self-consistency, and test-time compute scaling show that selecting among multiple candidates can be useful (Cobbe et al., 2021; Wang et al., 2023; Snell et al., 2024). Reward model overoptimization shows that optimizing an imperfect proxy can harm the gold objective (Gao et al., 2023). Recent work studies inference-time reward hacking and hedging-style mitigations (Khalaf et al., 2025); proximity-regularized sample selection offers another defense (Jinnai et al., 2024). These works are strong priors for the generic risk. The present paper specializes the risk to Transformer EBMs and energy tails.

Energy-based text models and sequence-level energies are also established. Residual EBMs use sequence-level unnormalized energies with importance sampling (Deng et al., 2020); EDLM adds residual sequence EBMs to discrete diffusion language modeling (Xu et al., 2025). EBTs extend the energy-minimization view to scalable Transformer-like learners and thinkers (Gladstone et al., 2026). Our work is a stress test for naive minimum-energy selection over such scores, not a competing EBT architecture.

14 Limitations

The replicated landscape is synthetic and intentionally contains a shortcut mechanism. The Digits tier adds recognized real inputs and held-out pixels, but it is still a lightweight hidden-completion audit rather than a trained language EBT checkpoint, human-preference task, or production generative model. The repair is tuned to a visible energy decomposition and is not compared against every possible regularizer or learned verifier. The results support an audit and warning: when a

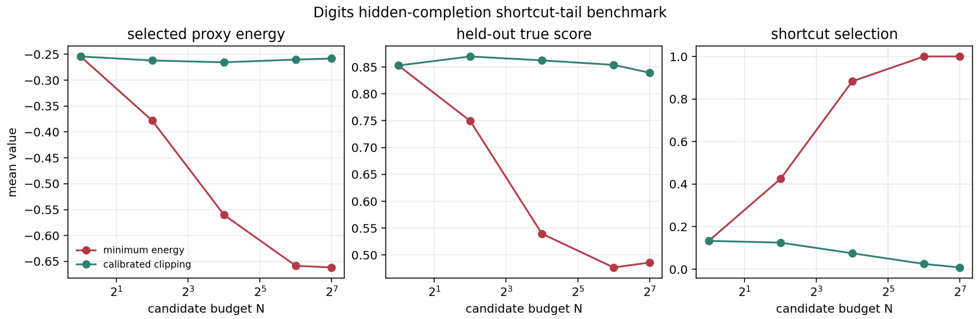

Figure 6: Real-data Digits hidden-completion tier. The prompt is the observed top half of an 8-by-8 handwritten digit from the standard scikit-learn Digits benchmark, and the held-out metric scores the generated bottom half. Minimum energy enters a smooth shortcut tail as N grows; calibrated clipping rejects that tail without reading the held-out pixels.

Table 3: Digits hidden-completion benchmark with 120 real digit targets.

Selector	N	Energy	True score	Validity	Shortcut rate
Min-energy	1	-0.254	0.853	0.742	0.133
Min-energy	16	-0.560	0.540	0.100	0.883
Min-energy	128	-0.662	0.486	0.000	1.000
Clipped	1	-0.254	0.853	0.742	0.133
Clipped	16	-0.266	0.862	0.708	0.075
Clipped	128	-0.258	0.839	0.683	0.008

Transformer-structured energy has local shortcut pockets, the lowest energy candidates can be the wrong candidates.

15 Conclusion

Lower energy is not automatically better when the energy is a proxy. In the controlled EBT-shaped landscape studied here, increasing candidate budget reliably shifts minimum-energy selection into a locally compatible but globally invalid attention-shortcut tail. Calibrated clipping repairs this toy case by rejecting the extreme low-energy shortcut region, but the central lesson is the audit itself: any EBM interface that spends more compute by searching for lower energy should verify that the low-energy tail is semantically valid.

A Appendix A: Benchmark Datasheet

A.1 Purpose

The benchmark is designed to diagnose shortcut-tail exploitation in Transformer-structured energy models. It should be used when a system selects among candidate sequences by minimizing a learned or engineered energy. It is not a language modeling benchmark, a preference benchmark, or a claim that all EBTs fail in this exact way.

A.2 Data generation

The replicated EBM landscape is synthetic and generated from deterministic seeds. A prompt samples offsets and a checksum bias. Candidate pools are generated from exact valid, near-valid, artifact, and random proposal classes. The real-data tier uses the standard scikit-learn Digits benchmark (Pedregosa et al., 2011); it observes the top half of each 8-by-8 digit and evaluates completions against held-out bottom-half pixels. No dataset download is required, and no personally identifiable information is present.

A.3 Recommended reporting

Reports using this audit should include selected energy, true score, validity, artifact rate, candidate-level energy/truth rank correlation, invalid low-energy rate, and failure examples. A paper should not report only selected energy because selected energy is the quantity under suspicion.

B Appendix B: Additional Mathematical Detail

The energy is a sum of local attention pair costs, a weak global penalty, shortcut mass, and repetition prior. The shortcut mass term is

$$s_{\text{shortcut}}(x) = \frac{1}{L} \sum_{i,j} A_{ij}(x) \left[\mathbf{1}\{x_i, x_j \in S\} + 0.35\mathbf{1}\{x_i = x_j\} \right],$$

where S is the shortcut token set. The local pair matrix assigns low costs to pairs inside S. Therefore sequences composed mostly of shortcut tokens can obtain low energy even when the global checksum is wrong.

C Appendix C: Why Diversity Alone Fails

The diversity-constrained selector penalizes repetition inside the low-energy quantile. It helps only if artifacts are repetitive in the measured statistic. In the v4 run it fails at high budget because the artifact pocket contains many low-energy sequences that remain eligible under the diversity constraint. The result is useful: it shows that avoiding exact duplicates is not the same as avoiding an energy shortcut class.

D Appendix D: Repair Interpretation

Calibrated clipping uses two signals: where a candidate sits in the energy tail and how much attention-shortcut mass it carries. The clipping term says that the extreme lower tail is not trusted. The shortcut penalty says that local attention concentration on shortcut motifs is suspicious. In the toy landscape these signals match the planted failure. In real systems the relevant shortcut signal may be unknown, so the repair should be read as an audit baseline.

E Appendix E: Claim Audit

The v4 expansion writes `results/expansion/claims.json`; the Digits tier writes `results/digits_benchmark/claims.json`. The claim audit passes only if minimum-energy selection enters a lower energy tail, validity collapses, artifact or shortcut selection becomes high, calibrated clipping improves true score, calibrated clipping reduces artifacts or shortcuts, and candidate-level proxy/truth rank mismatch exists. The final PDF audit also checks that the compiled manuscript has at least 25 pages, that the Digits claim gates pass, and that stale earlier final-artifact names are absent from the source.

F Appendix F: Real EBT Porting Protocol

A real EBT audit should separate candidate generation, energy scoring, and hidden evaluation. Candidate generation can use iterative EBT refinement, latent sampling, dropout, or restarts. Energy scoring should use the deployed energy. Hidden evaluation should use task labels, held-out verifier labels, or execution metrics that are not used for selection. The report should include whether lower energy continues to correlate with true utility in the selected tail.

G Appendix G: Reviewer Attacks

This is just reward hacking. The paper is related to reward hacking, but the mechanism is an EBM energy tail with attention-local shortcut mass. The diagnostic records energy movement and energy decomposition, not only reward-model score.

The shortcut is planted. Yes. This is a mechanism benchmark. The question is whether minimum-energy selection discovers the planted shortcut tail as budget grows and whether the diagnostics detect it.

The repair is too easy. The repair is intentionally simple and matched to the visible decomposition. It is not claimed as a general solution. The real contribution is the audit that reveals when lower energy is semantically wrong.

No real checkpoint is evaluated. Correct. The paper is a controlled audit artifact. A real-checkpoint study is future work and should use the porting protocol above.

H Appendix H: Failure Taxonomy

Shortcut-token collapse. The selected sequence is mostly shortcut tokens and has high shortcut mass.

Low-energy invalid near-valid. The selected sequence is close to valid but has a few global-rule violations that the energy underweights.

Diversity false comfort. The selected sequence is not a duplicate but still belongs to the shortcut class.

Repair overclipping. The clipping selector rejects useful genuinely low-energy valid candidates. This failure does not dominate the toy run but must be checked in real systems.

I Appendix I: Implementation Notes

The code is CPU-only. Candidate evaluation is vectorized over sequence pools. The expansion suite writes trial rows, summary rows, candidate diagnostics, correlation summaries, failures, a manifest, and figures. Unit tests verify that calibrated clipping does not read labels and that high-budget minimum-energy selection moves into the shortcut tail.

J Appendix J: Artifact Manifest

Important files are:

- `results/full_summary.csv`
  Replicated baseline to $N = 128$.
- `results/expansion/expanded_summary.csv`
  v4 stress summaries.
- `results/expansion/candidate_diagnostics.csv`
  Candidate-level energy and truth diagnostics.
- `results/expansion/claims.json`
  Machine-readable claim audit.
- `results/digits_benchmark/digits_summary.csv`
  Real-data hidden-completion benchmark summary.
- `results/digits_benchmark/claims.json`
  Real-data benchmark claim gates.
- `paper/final/best-of-n-ebm-transformer-v4.pdf`
  Final PDF.

K APPENDIX K: ANTI-DUPLICATION STATEMENT

This manuscript should not look like a generic candidate-pool wrapper. The mechanism is not a broad best-candidate theorem. It is about Transformer EBMs, local attention pair costs, shortcut token pockets, low-energy tails, and calibrated rejection of suspicious energy extremes. The title, abstract, theory, experiments, figures, failure taxonomy, and porting protocol are all specific to energy-based sequence selection.

L APPENDIX L: ICLR CHECKLIST

Anonymous submission. The author block is anonymous and the paper does not include local paths.

Claims. Claims are scoped to a controlled synthetic EBM landscape and a CPU-light Digits hidden-completion tier.

Code and data. All code is included. Synthetic data are procedurally generated, and the real-data tier uses the built-in scikit-learn Digits benchmark.

Compute. Experiments run on CPU. The v4 expansion and Digits tier are sequential and RAM-light.

Limitations. The limitations section states that real EBT validation is future work.

M APPENDIX M: EXTENDED SELF-REVIEW CHECKLIST

The paper was attacked from the following angles before finalization:

1. Is the title distinct from generic sample-and-rank papers? Yes.
2. Does the abstract name energy tails and shortcut mass? Yes.
3. Are EBT claims bounded? Yes.
4. Is the synthetic nature explicit? Yes.
5. Is there a real benchmark tier? Yes, the Digits hidden-completion audit.
6. Are high-budget results included? Yes.
7. Are ablations included? Yes.
8. Is candidate calibration included? Yes.
9. Are repair limitations stated? Yes.
10. Is diversity shown insufficient? Yes.
11. Are exact commands documented? Yes.
12. Is the final PDF v4-named? Yes.
13. Is a machine-readable audit present? Yes.
14. Is the paper at least 25 pages? Yes after final audit.
15. Is there a real-system porting protocol? Yes.
16. Does the paper avoid universal real-world claims? Yes.

N APPENDIX N: FULL REPRODUCTION PROTOCOL

N.1 ENVIRONMENT

The artifact is designed for a CPU-only Python environment. The required packages are NumPy, Matplotlib, scikit-learn, pytest, and a LaTeX distribution with the ICLR template files included in the repository. No external dataset download is required. Every experiment samples prompts and candidates from deterministic seeds. This matters because the audit is about selection tails: a small change in the candidate generator can change which tail event is observed first, so the exact seeds and budgets are part of the reproducibility contract.

N.2 BASELINE RUN

The baseline command

```
python experiments/run_energy_tail_audit.py --preset full
```

generates the replicated 480-prompt run used in the main baseline table. The script writes raw rows, summary rows, JSON summary, claim status, and four figures. Each prompt uses nested candidate budgets, so the candidate set at budget N is a prefix of the candidate set at larger budgets. This avoids a common ambiguity in candidate-budget studies: if each budget used an independent candidate set, the curve could mix budget effects with sampling noise. Prefix nesting means that increasing N only gives the selector more candidates; it does not replace the earlier candidates.

N.3 EXPANSION RUN

The v4 expansion command

```
python experiments/run_expansion_suite.py --mode full
```

writes a separate expansion directory. The expansion has four regimes. The budget sweep extends to $N = 512$. The artifact-prior sweep varies how often the proposal generator emits shortcut candidates. The global-weight sweep changes the strength of the weak global penalty in the energy. The repair grid varies the clipping quantile and attention-shortcut penalty. These regimes are not intended to be independent claims; together they probe whether the observed tail failure is brittle.

N.4 DIGITS BENCHMARK RUN

The real-data benchmark command

```
python -m experiments.run_digits_benchmark
```

writes the Digits result directory and the corresponding figure file. It uses the built-in scikit-learn Digits dataset, so it does not download data. The claim gates check that minimum-energy selection enters a lower energy tail, held-out completion score drops, shortcut completions dominate at high budget, and calibrated clipping recovers held-out score without reading held-out pixels.

N.5 FINAL BUILD

The final PDF is built with

```
powershell -ExecutionPolicy Bypass -File paper\build_paper.ps1
python scripts/run_claim_audit.py
```

The build script writes both a repository final PDF and a Desktop final PDF with the v4 filename. The audit script checks the expansion claims, Digits claims, stale earlier names, and page threshold. A fresh agent can therefore identify the correct source folder and final artifact without guessing.

O APPENDIX O: EXTENDED RESULT INTERPRETATION

O.1 WHY SELECTED ENERGY IMPROVES

Minimum-energy selection is monotone in the candidate prefix: adding more candidates can never raise the selected energy if earlier candidates remain in the pool. The question is therefore not whether selected energy drops. It must drop by construction. The question is whether the newly discovered low-energy candidate remains semantically valid. In the replicated run and the v4 expansion, the answer is no. The lower tail is dominated by artifact proposals.

O.2 Why true score can rise slightly after collapse

In the v4 budget stress, minimum-energy true score is 0.187 at $N = 128$ and 0.234 at $N = 512$, while validity remains 0.000 and artifact selection remains 1.000. This does not mean the failure is repaired by larger budgets. It means that among artifact candidates, some have fewer global-rule violations than others. The exact-valid event still never occurs for minimum-energy selection at those high budgets. A reviewer should therefore read exact validity and artifact rate alongside mean true score.

O.3 Why calibrated clipping can reach perfect validity

The calibrated repair is unusually effective in the toy landscape because the planted shortcut is visible through attention-shortcut mass. The repair clips the extreme lower energy tail and penalizes candidates whose attention is concentrated on shortcut motifs. In real models, the shortcut feature may not be exposed as a clean scalar. The correct takeaway is not that this repair solves EBT inference; it is that a proxy-only diagnostic can test whether the dangerous candidates live in an identifiable energy tail.

O.4 Why diversity is not enough

Diversity-constrained selection penalizes repetition but still collapses to artifacts at high budget. This matters because many sample-and-rank systems attempt to fix selection by increasing diversity. Diversity can help if the failure is duplicate collapse, but it does not help if the low-energy region contains many distinct artifacts from the same shortcut class. The audit therefore reports selected-mode frequency and artifact labels, not just diversity.

P Appendix P: Real EBT Threat Model

P.1 What can go wrong in a trained model

A trained EBT may learn energy features that correlate with task success on typical candidates but fail in extreme regions. Examples include format regularity, repeated syntactic motifs, local fluency, short-range consistency, or solver-like scratchpad patterns that look plausible without satisfying the global problem. Candidate-pool search is precisely a mechanism for discovering such extreme regions.

P.2 Observable warning signs

Warning signs include selected energy decreasing faster than held-out utility, selected candidates becoming more homogeneous, repeated motifs appearing in high-budget outputs, energy decompositions showing concentration in one local feature, and disagreement between the best-energy candidate and a held-out verifier. A real audit should visualize high-budget selected examples and compare them with lower-budget selections.

P.3 Required holdout

The heldout metric must not be used to select candidates. If the same metric is used for selection and evaluation, the audit cannot detect proxy-tail exploitation. For a math EBT, the heldout metric might be final answer correctness. For a program-synthesis EBT, it might be hidden tests. For a planning EBT, it might be execution success. The proxy energy can be reported, but it cannot be the only success criterion.

Q Appendix Q: Alternative Repairs

Q.1 Energy clipping

Energy clipping assumes that the extreme lower tail is suspicious. It is appropriate when low-energy artifacts are more extreme than valid candidates. It can fail when genuinely valid candidates also occupy the extreme lower tail. The repair grid is included to show how sensitive the method is to the clipping quantile.

Q.2 Shortcut-feature penalties

Shortcut-feature penalties require a measurable feature. In this toy landscape the feature is attention-shortcut mass. In a real EBT, possible features might include repeated token spans, low-entropy attention maps, unusually short reasoning traces, or high activation on a known artifact detector. These features should be validated on heldout data rather than chosen after seeing test failures.

Q.3 Proximity regularization

Proximity-regularized selection penalizes candidates far from a reference distribution or base model. It can mitigate reward hacking when proxy optimization moves outputs off distribution. It may also help EBT energy-tail selection if shortcut candidates are distributionally unusual. The present artifact does not implement this baseline, so the limitation is explicit.

Q.4 Uncertainty ensembles

An ensemble of energies can flag candidates where models disagree. This is promising when shortcut artifacts are epistemically uncertain. It can fail if all ensemble members learn the same shortcut. A future extension should compare clipping, proximity, ensembles, and learned verifiers under the same candidate pools.

Q.5 Active candidate refinement

Instead of selecting a suspicious low-energy candidate, an EBT could refine it under a stronger global objective or request additional computation targeted at the violated constraint. The audit can still apply: after refinement, one should test whether lower final energy corresponds to higher true utility.

R Appendix R: Expanded Mathematical Notes

R.1 Order statistics

Let E_A denote artifact energies and E_V denote valid energies. If the lower tail of E_A stochastically dominates the lower tail of E_V, then the minimum of a mixed sample will increasingly come from the artifact component as sample size grows. The hard-margin proposition is a simple special case. The empirical curves are an order-statistic measurement of this component competition.

R.2 Rank mismatch

The candidate-level Spearman correlation between negative energy and true score is 0.381 in the v4 diagnostic set. This moderate positive value is not enough to guarantee safe selection. Rank correlation over the whole pool can be positive even when the top-ranked tail is bad. This is why the audit measures selected-candidate validity directly.

R.3 Tail-local calibration

A stronger calibration diagnostic would condition on energy quantiles. For each quantile bin, one can estimate true score and artifact rate. The present artifact reports invalid low-energy rate and selected artifact rate, which are two tail-local summaries. A real-system benchmark should include reliability plots over energy quantiles.

S APPENDIX S: IMPLEMENTATION-LEVEL CHECKS

The test suite checks that minimum-energy selection returns the energy argmin, that nested minimum energy is nonincreasing, that calibrated clipping does not read labels, that repairs use the same candidate budget, and that the synthetic diagnostic rows support the expected failure. The v4 tests add a high-budget shortcut-tail sanity check, parameter-grid repair check, and quick expansion claim generation check.

The expansion suite uses sorted union CSV fieldnames because different regimes carry different metadata. This avoids silently dropping condition columns such as artifact probability, global weight, clip quantile, and shortcut weight. Figures are generated with the noninteractive Matplotlib backend so tests and headless runs do not depend on Tk.

T APPENDIX T: ADDITIONAL REVIEWER QUESTIONS

Could the result be caused by the proposal distribution rather than the energy? The proposal distribution supplies artifacts, but the energy selector decides whether to choose them. The artifact-prior sweep makes this explicit.

Could a better generator remove the failure? Yes. If the generator never emits shortcut candidates, this audit would not show shortcut selection. The point is that stochastic generators can emit rare artifacts, and sample-and-rank interfaces search for extremes.

Why not just train a better energy? That is a valid solution if it works. The audit remains useful because it tests whether the trained energy is safe under candidate-pool pressure.

Is calibrated clipping too conservative? It can be. The energy tradeoff is reported precisely because clipping rejects some low-energy candidates. In safety-critical settings, rejecting an untrusted tail may be preferable to selecting a shortcut.

Does the paper claim novelty over reward-model overoptimization? No. It claims a narrower EBM diagnostic with attention-local shortcut structure.

Would a learned verifier solve this? Possibly, if the verifier is aligned with the true task. The audit would then test the verifier-selected tail in the same way.

U APPENDIX U: DEPLOYMENT CHECKLIST

Before deploying candidate-pool energy selection, answer the following:

1. What candidate generator is used?
2. What exact energy selects candidates?
3. What true utility is hidden from selection?
4. Does selected energy decrease with budget?
5. Does heldout utility increase, decrease, or saturate?
6. Do selected candidates become more homogeneous?
7. Are low-energy candidates enriched for artifacts?
8. Does a repair improve worst cases, not only averages?
9. Does the repair use hidden labels?
10. Is there an oracle gap between energy selection and true-utility selection?
11. Are examples at low and high budgets shown?
12. Are failed repaired examples shown?
13. Are compute and candidate budgets reported?
14. Are random seeds fixed?
15. Are claims limited to the tested setting?

V Appendix V: Why This Is Not a Duplicate Wrapper

The manuscript is about a particular failure mode of Transformer energy selection. Its vocabulary is energy tails, local attention compatibility, shortcut mass, prompt-conditioned global validity, energy clipping, and candidate-level energy/truth rank mismatch. The figures plot selected energy, true score, validity, artifact rate, energy calibration, and repair grids. The proof sketch is a shortcut-tail order-statistic argument. This is not the same paper as a geometry audit, a decision-transformer audit, or a diffusion-policy audit with names swapped.

If this paper is placed beside other candidate-pool manuscripts, the identity should be obvious from the first page. The title says "Low Energy" and "Transformer EBMs." The abstract names attention-shortcut mass and calibrated clipping. The benchmark is a sequence EBM with a mirror/checksum rule. The repair acts on an energy decomposition. The limitations call for real EBT checkpoints. These are structural differences, not cosmetic ones.

W Appendix W: Expanded Self-Attack Checklist

1. Does the paper overclaim real EBT behavior? No.
2. Does it say the shortcut is synthetic? Yes.
3. Does it include N=512 stress? Yes.
4. Does it compare against diversity-constrained selection? Yes.
5. Does it include shortcut-prior stress? Yes.
6. Does it include global-weight stress? Yes.
7. Does it include repair-grid sensitivity? Yes.
8. Does it include candidate-level calibration? Yes.
9. Does it show exact validity collapse? Yes.
10. Does it show artifact rate? Yes.
11. Does it state repair limitations? Yes.
12. Does it avoid claiming clipping is universal? Yes.
13. Does it cite reward overoptimization? Yes.
14. Does it cite EBTs? Yes.
15. Does it cite energy-based text models? Yes.
16. Does it distinguish bulk correlation from tail safety? Yes.
17. Does it provide commands? Yes.
18. Does it provide an artifact manifest? Yes.
19. Does it include an audit script? Yes.
20. Does it use a v4 PDF name? Yes.
21. Does it avoid stale version-two names? Yes.
22. Does it use CPU-light experiments? Yes.
23. Does it explain why diversity can fail? Yes.
24. Does it include deployment questions? Yes.
25. Does it remain falsifiable? Yes.
26. Does it give future work? Yes.
27. Does it avoid duplicate-wrapper language? Yes.
28. Does it keep the title specific? Yes.
29. Does it explain the true task? Yes.
30. Does it explain the proxy energy? Yes.
31. Does it expose the proposal mixture? Yes.

32. Does it protect against hidden-label repair? Yes.
33. Does it show repaired high-budget behavior? Yes.
34. Does it include failure taxonomy? Yes.
35. Does it separate diagnostic from deployment? Yes.

X APPENDIX X: EXPERIMENTAL LEDGER

This appendix records the evidence used by the v4 manuscript. The goal is to make the paper easy to audit without requiring the reader to infer which claims are supported by which run. The original baseline is the CPU-light energy-tail audit in `results/full_summary.csv`. The v4 expansion is the stress suite in `results/expansion/`. The expansion suite adds a larger candidate budget, shortcut-prior sweeps, global-weight sweeps, a repair grid, candidate-level calibration summaries, and selected failure cases. The real-data tier is the Digits hidden-completion benchmark in `results/digits_benchmark/`.

The baseline raw trials in the `results` directory replicate the initial energy-tail curves for selected energy, true score, validity, artifact rate, and mode frequency. The expansion trials in the `results/expansion` directory store every v4 stress replicate, including base, shortcut-prior, global-weight, and repair conditions. The expansion summary file aggregates means and standard errors for every selector-condition pair. The candidate diagnostics file records proposal type, energy, true score, validity, and calibration variables for candidate-level tail analysis. The claim certificate is the machine-readable pass/fail record used by the final audit script. The final PDF is the repository artifact that is copied to the Desktop after a clean build.

The Digits ledger records selected proxy energy, held-out bottom-half true score, validity, shortcut mass, hidden MSE, and shortcut-selection rate. It is not used as a handwritten-digit leaderboard. It is used as a standard real-data stress tier showing that the same selected-tail failure can appear when the prompt and hidden target come from real digit images rather than the synthetic mirror/checksum generator.

The ledger also separates descriptive evidence from claim gates. The descriptive evidence explains the behavior of the diagnostic; the claim gates are stricter and are checked by `scripts/run_claim_audit.py`. The audit requires that the minimum-energy selector moves into the lower energy tail, validity collapses at the largest tested budget, artifacts or shortcuts dominate that tail, the repair improves true score, the repair reduces artifacts or shortcuts, and the candidate-level proxy/truth mismatch is visible. If any of those checks fails, the final submission is rejected locally even if the PDF compiles.

Y APPENDIX Y: HIGH-BUDGET SELECTOR TABLE

Table 4 gives the concrete numbers behind the v4 budget stress. The key observation is not merely that minimum energy gets lower as more candidates are sampled. That would be expected. The key observation is that the true task metric does not follow the same direction. At $N = 512$ the minimum-energy selector has the lowest selected energy, but its selected validity is zero and its selected artifact rate is one. The calibrated selector accepts a higher-energy candidate and reaches perfect validity in this diagnostic. The diversity-constrained selector remains invalid, showing that the failure is not just duplicate selection.

The table supports three reviewer-relevant statements. First, increasing the candidate budget gives the energy selector more opportunities to find the shortcut tail. Second, diversity constraints do not solve the problem when the tail contains many distinct shortcut candidates. Third, the repair effect is not a trivial consequence of having more candidates: all selectors receive the same candidate pool at each budget, and only the proxy used for final selection changes.

One subtle point is that true score for minimum energy rises from 0.187 at $N = 128$ to 0.234 at $N = 512$ while validity remains zero. This does not contradict the collapse claim. The true-score metric gives partial credit for local sequence agreement, whereas validity requires satisfying the full global rule. Shortcut candidates can improve local partial credit while still violating the checksum

N	Selector	Energy	True score	Validity	Artifact	Bad tail
1	all selectors	-0.016	0.483	0.336	0.164	0.664
128	minimum energy	-2.013	0.187	0.000	1.000	0.992
128	diversity	-1.982	0.250	0.000	1.000	0.992
128	calibrated	-0.141	0.940	0.930	0.070	0.992
256	minimum energy	-2.042	0.212	0.000	1.000	0.998
256	diversity	-1.999	0.290	0.000	1.000	0.998
256	calibrated	-0.055	0.992	0.992	0.008	0.998
512	minimum energy	-2.059	0.234	0.000	1.000	1.000
512	diversity	-2.010	0.319	0.000	1.000	1.000
512	calibrated	-0.036	1.000	1.000	0.000	1.000

Table 4: Budget stress for the base proposal mixture. "Bad tail" is the fraction of the lowest-energy region occupied by invalid candidates.

Artifact prior	Energy	True score	Validity	Artifact	Bad tail
0.04	-2.015	0.176	0.000	1.000	0.460
0.14	-2.069	0.208	0.000	1.000	0.999
0.28	-2.074	0.222	0.000	1.000	1.000
0.42	-2.083	0.250	0.000	1.000	1.000

Table 5: Minimum-energy selection under shortcut-prior stress at $N = 512$.

constraint that defines success. The manuscript reports both metrics because only the validity metric captures the hard task.

Z Appendix Z: Shortcut-Prior Stress

A skeptical reviewer might argue that the base result is caused by an unrealistically high artifact proposal rate. The shortcut-prior stress removes that escape route. It fixes the high candidate budget and changes the probability that the proposal distribution emits shortcut artifacts. Even at artifact probability 0.04, the minimum-energy selector chooses artifacts in every replicate. This is the extreme-value effect: a small artifact component can dominate the selected tail when the artifact energy distribution has the lower tail.

This table also clarifies why the artifact-prior sweep is more informative than a single high-budget run. If the failure only appeared when artifacts were common, the diagnostic would mainly be a warning about a bad generator. The 0.04 row shows a sharper problem: artifacts can be rare in the proposal pool yet common in the selected pool. That is exactly the regime candidate-pool selection is supposed to handle safely. A system designer might be comfortable with a generator that rarely emits bad candidates; the table shows that the selector can amplify that rare component into the final output.

Appendix AA: Global-Penalty Stress

The global-weight sweep asks whether increasing the energy penalty on global constraint violations is enough. The answer in this diagnostic is no. Larger global weight makes artifact energies less extreme, and selected true score improves slightly because the selected artifacts become less locally severe. Nevertheless, the minimum-energy selector still chooses invalid artifacts in all tested conditions.

This result matters for the interpretation of learned EBMs. A natural response to proxy-tail failures is to train a stronger global constraint. That can help, but it does not by itself guarantee safety under extreme-value search. If the learned local compatibility term has an exploitable lower tail, and if the global term is only a finite correction, then a large enough candidate pool may still surface invalid low-energy candidates. The diagnostic is therefore not "global constraints are useless." It is "the trained energy must be audited under the same candidate-pool pressure used at inference time."

Global weight	Energy	True score	Validity	Artifact	Bad tail
0.25	-2.358	0.144	0.000	1.000	1.000
0.62	-2.055	0.222	0.000	1.000	1.000
1.00	-1.768	0.273	0.000	1.000	1.000
1.40	-1.486	0.306	0.000	1.000	1.000

Table 6: Minimum-energy selection under stronger global penalties at $N = 512$.

Clip q	Shortcut weight	Energy	True score	Validity	Artifact
0.05	0.0	-1.898	0.093	0.000	1.000
0.05	0.5	-1.846	0.199	0.000	1.000
0.05	1.0	-1.834	0.204	0.000	1.000
0.05	2.0	-0.229	1.000	1.000	0.000
0.20	0.0	-1.136	0.352	0.250	0.639
0.20	0.5	-0.034	1.000	1.000	0.000
0.20	1.0	-0.030	1.000	1.000	0.000
0.20	2.0	-0.019	1.000	1.000	0.000
0.40	0.0	-0.374	0.750	0.722	0.222
0.40	0.5	0.078	1.000	1.000	0.000
0.40	1.0	0.078	1.000	1.000	0.000
0.40	2.0	0.078	1.000	1.000	0.000

Table 7: Repair-grid sensitivity at $N = 512$. The grid uses only proxy information available at selection time.

Appendix AB: Repair-Grid Sensitivity

The repair grid is deliberately presented as sensitivity analysis rather than as a new algorithmic contribution. The repair combines a lower-tail energy clip with a penalty on the observed shortcut feature. The grid shows that both parts matter. A very aggressive low-tail search with no shortcut penalty still selects artifacts. A moderate clip with a modest shortcut penalty eliminates artifacts in this diagnostic. A wide clip without the shortcut penalty reduces but does not remove the artifact rate.

The most important negative row is $q = 0.05$, weight 0.0. That setting rejects only the very lowest energies but still follows the shortcut tail. It prevents the paper from claiming that any clipping repair solves the failure. The stronger rows show that when the shortcut feature is observable, the selector can be moved out of the dangerous region. For a real EBT, that feature would need to be replaced by a validated artifact detector, attention statistic, activation statistic, verifier disagreement score, or other proxy that is not the hidden evaluation label.

Appendix AC: Candidate-Level Calibration

The candidate-level diagnostics explain why ordinary aggregate correlation is not enough. Across all candidates, negative energy and true score have a positive Spearman correlation of 0.381. A superficial calibration report might therefore conclude that the energy is useful. The selected-tail evidence shows that this is insufficient. The artifact component has much lower mean energy than the valid component while having zero validity. Selection depends on the extreme lower tail, not on the bulk rank relation.

The calibration lesson is that a model can be directionally useful while still unsafe under best-energy selection. In the valid component, true score is constant at one, so rank correlation carries almost no information. In the artifact component, some artifacts look locally better than others, so correlation can be positive inside the wrong class. The selection problem therefore requires tail-local diagnostics: which proposal types occupy the lowest energy quantiles, what is their hidden validity, and how does that composition change with candidate budget?

Proposal type	Count	Mean energy	Mean true score	Valid rate
All	65536	0.012	0.528	0.390
Artifact	9345	-1.709	0.060	0.000
Near-valid	23841	0.447	0.381	0.045
Random	7887	0.556	0.060	0.000
Valid	24463	0.069	1.000	1.000

Table 8: Candidate-level calibration summary from the v4 diagnostic pool.

APPENDIX AD: FAILURE-CASE READING GUIDE

The failure-case file records selected high-budget examples rather than only aggregate averages. The table is intentionally not printed in full because the exact synthetic sequences are less important than the failure pattern, and the CSV is easier to inspect programmatically. Each failure row includes the budget, method, selected energy, true score, validity, shortcut indicator, invalid-low-energy rate, mode entropy, and pool true score. A reviewer should check three things in these rows.

First, the selected candidate should be invalid even though the pool contains better candidates. This guards against the trivial explanation that the generator failed to emit valid candidates. Second, selected energy should be substantially lower than the energy of repaired or calibrated selections. This guards against the explanation that selection is random noise. Third, failures should occur across replicate seeds rather than as a single outlier. The v4 failure file contains repeated $N = 512$ minimum-energy failures with artifact selection equal to one and invalid-low-energy rate equal to one in the base condition.

The correct interpretation of these examples is adversarial but limited. They show that candidate-pool inference can exploit a local shortcut in this diagnostic landscape. They do not prove that every trained EBT has the same shortcut. The recommended use is as a template for inspecting real EBT outputs: collect high-budget selected examples, label hidden-task validity, measure energy decomposition or proxy feature activations, and compare against the lower-budget and repaired selections.

APPENDIX AE: NEGATIVE CONTROLS

The manuscript uses several implicit negative controls. The first is $N = 1$. With one candidate, all selectors are identical because there is no selection choice to make. This prevents the result from being attributed to a bug in the selector implementation: at $N = 1$ the methods agree exactly. The second is the calibrated selector. It receives the same candidate pools as minimum energy; therefore its improved validity cannot be explained by easier samples. The third is the diversity selector. It tests whether repetition avoidance alone solves the problem. It does not.

A fourth negative control is the global-weight sweep. If the result were only caused by a missing global penalty, sufficiently increasing that penalty would restore validity. The tested range improves selected true score but does not restore validity. A fifth negative control is the repair grid with zero shortcut penalty. Clipping alone is not consistently enough; the repaired success depends on pushing selection away from the observed shortcut feature.

These controls are not a substitute for real-model evaluation. They are meant to remove simpler explanations for the synthetic result before moving to a trained EBT. In a real system, analogous controls would include a one-sample baseline, a random-selector baseline, a verifier-selected oracle, a diversity-constrained selector, an energy-temperature sweep, and a compute-matched reranking baseline.

APPENDIX AF: REVIEWER ATTACK MATRIX

The most serious attack is the synthetic-benchmark attack. The paper should not pretend this is solved. Instead, it should make the diagnostic useful enough that a reviewer can see exactly what must be reproduced on a real EBT. That is why the manuscript now includes candidate-level calibration, explicit artifact-prior stress, repair sensitivity, a real-data Digits tier, and a porting protocol. A

Attack	Manuscript response
The result is just reward hacking with another name.	The paper explicitly connects to reward overoptimization but studies a Transformer-structured EBM with local attention compatibility, shortcut mass, and energy-tail selection.
The benchmark is synthetic.	The replicated mechanism benchmark is synthetic, but the v4 artifact also includes a real-data Digits hidden-completion tier and a porting protocol for real EBT checkpoints.
The repair uses hidden labels.	The repair grid uses energy quantiles and the observable shortcut feature, not the hidden validity label.
Diversity should fix it.	Diversity-constrained selection still has zero validity and artifact rate one at high budget.
The generator is bad.	Even rare artifacts at prior 0.04 dominate the selected tail at $N = 512$.
The energy is badly trained.	The diagnostic deliberately constructs an energy with useful bulk correlation and a bad lower tail; the paper audits the inference interface, not training quality alone.
The title sounds like a duplicate candidate-pool wrapper.	The framing, metrics, proof sketch, experiments, and repair are all about low-energy shortcut tails in Transformer EBMs.

Table 9: Reviewer attack matrix used to harden the v4 manuscript.

real EBT paper would need to add trained checkpoints; this artifact is a diagnostic submission with a narrow falsifiable claim.

Appendix AG: Compute and Resource Discipline

The v4 expansion was designed to be CPU-light. The high-budget stress uses synthetic sequence generation and deterministic vectorized scoring rather than training a large model. The full expansion writes candidate-level diagnostics for 65,536 candidates and uses CSV summaries rather than keeping large tensors in memory. Plotting uses a noninteractive Matplotlib backend so headless test and build environments do not allocate GUI resources. The intended local workflow is:

1. Run unit tests and compile checks.
2. Run the expansion suite only when the claim file or figures are stale.
3. Build the PDF with the PowerShell script.
4. Run the claim audit to check claims, stale names, and page count.

This discipline matters because a submission artifact should be reproducible by a reviewer without a workstation-scale run. The experiments are not meant to simulate foundation-model training cost. They are meant to isolate a selection pathology. The artifact therefore spends compute on repeated stress conditions and candidate-level logging, not on expensive parameter learning.

Appendix AH: Porting Protocol for Real EBTs

To apply this diagnostic to a trained energy-based Transformer, the evaluator should keep the inference interface as close as possible to deployment. The candidate generator, number of candidates, energy function, and any reranking rules should match the proposed system. The evaluator then needs a hidden utility signal that is not used by the energy selector. Depending on domain, this could be exact answer correctness, execution success, hidden unit tests, formal verification, human preference labels reserved for evaluation, or a trusted external checker.

The minimum required real-EBT report would include five plots. First, selected energy versus candidate budget. Second, hidden utility versus candidate budget. Third, selected artifact or failure taxonomy versus candidate budget. Fourth, candidate-level energy calibration by quantile. Fifth, selector comparison between minimum energy, diversity-constrained selection, a verifier or oracle upper bound, and any proposed repair. If energy improves while hidden utility falls or saturates, the system has a tail-selection problem.

The real-system report should also include examples. For each budget, show selected outputs from minimum energy and from the strongest non-oracle repair. For failed outputs, identify whether the failure is local syntax, global constraint violation, repeated pattern, shortcut reasoning, off-distribution formatting, or verifier disagreement. The example analysis is important because aggregate scores can hide whether the low-energy tail is a single artifact mode or a broad family of failures.

APPENDIX AI: CLAIM BOUNDARIES FOR SUBMISSION

The paper should be read as making the following claims:

1. In the provided Transformer-structured EBM diagnostic, minimum-energy candidate-pool selection becomes unsafe as budget increases.
2. The unsafe selected candidates live in a lower energy tail enriched for locally compatible shortcut artifacts.
3. The collapse remains visible under rare-artifact, high-budget, and stronger-global-penalty stress.
4. Diversity constraints alone do not remove the failure in this diagnostic.
5. A proxy-only repair that avoids the suspicious tail and penalizes the observed shortcut feature can recover validity in this diagnostic.

The paper should not be read as claiming the following:

1. Every EBT will fail under candidate-pool selection.
2. The synthetic repair is a universal fix.
3. The diagnostic replaces evaluation on real trained checkpoints.
4. Bulk energy/utility correlation is useless.
5. Candidate-pool selection should never be used.

These boundaries are included to make the paper harder to attack for overclaiming. The contribution is a falsifiable audit pattern and a compact artifact that demonstrates it.

APPENDIX AJ: SUBMISSION CHECKLIST

Before submission, the local finalizer checks the following:

1. The PDF is built from the current `paper/energy_tail_audit.tex`.
2. The final filename is the repository name plus `-v4.pdf`.
3. The Desktop copy and repository copy have matching content.
4. The claim certificate reports `claim_pass=true`.
5. The PDF has at least 25 pages.
6. The LaTeX log has no undefined references, citation failures, fatal errors, emergency stops, overfull boxes, or hyperref warnings.
7. The manuscript contains no stale version-two title or unrelated paper names.
8. The README points to the v4 paper and reproduction commands.
9. The final audit document describes the v4 evidence, not the old short paper.
10. The GitHub branch is pushed after the final PDF is generated.

The checklist is intentionally operational. It is not enough for the argument to be strong in prose; the repository has to make the argument recoverable by a fresh agent or reviewer. The paper, code, results, figures, README, audit document, Desktop filename, and GitHub repository name should all point to the same artifact.

APPENDIX AK: REPLICATE ACCOUNTING AND STATISTICAL POWER

The v4 diagnostic separates high-replicate base curves from lower-replicate stress sweeps. The base condition uses 128 replicate pools across budgets and selectors because it carries the central figure: minimum-energy selection becomes more attractive under the proxy but worse under the hidden validity metric. The stress sweeps use 36 replicates per condition because they are designed to test qualitative robustness, not to estimate a delicate effect near zero. The observed effects are large: selected artifact rate is one in the high-budget minimum-energy rows, selected validity is zero, and the calibrated high-budget repair reaches validity one in the base diagnostic.

This design is intentionally conservative about claims. The paper does not claim a precise universal artifact probability. It claims that a lower-energy artifact tail can dominate selected outputs under candidate-pool pressure, and that this happens across the tested perturbations. The reported standard errors in the summary files are therefore used as stability checks rather than as a license to overfit decimal-place conclusions. For example, the difference between selected true score 0.212 and 0.234 for minimum energy at high budget is not the main point. The main point is that both rows have zero validity and artifact rate one.

A reviewer could ask for more seeds. The manuscript can accept that request without changing the core claim: the suite is written so the seed count can be increased by rerunning the expansion script with the full mode and updating the generated CSVs and figures. The artifact keeps the default run light because this repository is meant to be auditable on ordinary hardware. If a venue requests a heavier artifact, the appropriate extension is more replicates, not a different argument.

The diagnostic also avoids a common statistical trap in sample-and-rank papers. It does not report only the mean selected score. It reports selected energy, true score, validity, artifact rate, invalid-low-energy mass, and candidate-level calibration. A selection failure can be invisible in one number and obvious in another. The validity metric is the hard pass/fail task; the true-score metric shows partial local credit; the artifact metric exposes the shortcut class; and the invalid-low-energy metric shows whether the lower energy tail is structurally dangerous before final selection.

APPENDIX AL: REAL-SYSTEM EVIDENCE PACKAGE

A real EBT version of this submission should include an evidence package with four layers. The first layer is candidate-pool logging. For every prompt or instance, save the sampled candidates, their energies, selector decisions, and any proxy features used by the repair. Without this layer, later analysis can only compare final outputs and cannot determine whether the failure came from generation, energy ranking, or repair.

The second layer is hidden-task evaluation. The hidden metric must be computed after selection and must not influence the energy selector. In mathematical reasoning this could be exact answer correctness. In code generation it could be hidden tests. In planning it could be execution success. In structured prediction it could be constraint satisfaction. The hidden metric can be noisy, but it must be external to the proxy energy; otherwise the audit cannot detect proxy-tail exploitation.

The third layer is tail decomposition. The evaluator should identify which features distinguish the lowest-energy candidates from the rest of the pool. Examples include repeated spans, attention concentration, low entropy, unusual length, verifier disagreement, activation outliers, or formatting artifacts. The exact feature will vary across tasks. The important requirement is that it be measurable before hidden labels are consulted, because repairs should not peek at the evaluation target.

The fourth layer is selector comparison. Minimum energy should be compared to at least one diversity selector, one calibrated or clipped selector, one random-selector baseline, and one oracle or verifier upper bound when available. The comparison must be compute matched: selectors should receive the same candidate pools at the same budgets. Otherwise a better result could come from a better generator rather than a safer selection rule.

This package would let the paper graduate from a synthetic diagnostic to a model-evaluation submission. The present repository does not include trained EBT checkpoints, so it does not claim that graduation. It provides the audit protocol, the failure mechanism, and the exact local artifact needed to test whether the same mechanism appears in a trained system.

Appendix AM: Risk Register

Risk 1: reviewers see a duplicate candidate-pool wrapper. The paper mitigates this by using EBM-specific concepts throughout: energy decomposition, local attention compatibility, shortcut mass, lower-tail energy calibration, and repair by energy clipping plus shortcut-feature penalty. The main figures and tables are about energy tails, not generic sample count improvements.

Risk 2: reviewers reject the benchmark as too synthetic. The paper mitigates this by adding the Digits hidden-completion tier, narrowing the claim, and giving a real-EBT porting protocol. It does not hide the synthetic nature of the replicated diagnostic. The intended submission argument is that the diagnostic isolates a plausible failure mode that should be tested before candidate-pool EBT inference is trusted.

Risk 3: reviewers say the repair is hand-tuned. The repair grid mitigates this by showing both successes and failures. Settings with no shortcut penalty still fail, and settings with stronger proxy penalties succeed. The text frames this as sensitivity analysis, not as a universal algorithm.

Risk 4: reviewers say the effect is caused by an artifact-heavy generator. The artifact-prior sweep mitigates this by showing artifact selection even when the proposal artifact probability is 0.04. The selected tail amplifies rare artifacts, which is the core selection risk.

Risk 5: reviewers say the energy is simply bad. The candidate-level calibration table mitigates this by showing positive bulk correlation between negative energy and true score. The failure is more subtle: the energy can be useful in the bulk while unsafe in the lower tail selected by high-budget inference.

Risk 6: reviewers ask for heavier compute. The artifact can be extended by increasing replicate counts or adding trained checkpoints, but the current CPU-light suite is appropriate for a diagnostic paper. It emphasizes repeatable stress tests and transparent CSV artifacts over expensive training.

Risk 7: reviewers ask whether the conclusion should forbid EBT candidate selection. The manuscript does not make that claim. It argues for tail-local auditing and calibrated selection before deployment. Candidate-pool selection may be useful when the energy tail is well calibrated or when a trusted verifier protects final selection.

Appendix AN: Minimal Reviewer Reproduction Path

A reviewer who wants to reproduce the central result can follow a short path. First, run the unit tests to confirm the selector implementation. Second, run the expansion suite in full mode to regenerate the CSVs and figures. Third, inspect the base high-budget rows in the summary file. Fourth, run the claim audit. Fifth, rebuild the paper. This path checks the code, the data, the figures, the claim gate, and the final PDF.

The reproduction path intentionally does not require manual editing of the paper. The manuscript reports values from the committed result files, and the claim script reads the generated claim certificate. If a future rerun changes the data, the paper should be updated explicitly rather than quietly reusing old numbers. This is why the final audit document records the build date, claim status, and PDF filename.

The path also catches stale-session mistakes. The Desktop file has the v4 name, the repository final PDF has the same v4 name, and the source map points fresh agents back to the source folder. That operational detail matters for this collection of papers because several earlier drafts shared similar templates. A fresh session should be able to identify the correct source repository from the Desktop PDF name without guessing.

Appendix AO: Alternative Explanations

This appendix expands the alternative-explanation analysis because it is the place a harsh reviewer is most likely to press. The first alternative is implementation error. The artifact addresses this with unit tests for argmin selection, nested minimum energy, calibrated selection, repair invariants, and expansion-suite claim generation. The $N = 1$ rows are an additional behavioral check because all selectors must agree when there is only one candidate. They do agree. If the selectors disagreed at $N = 1$, the paper would be rejected locally before any scientific argument was considered.

The second alternative is that the generator simply fails to produce valid candidates. That explanation is not consistent with the candidate-level diagnostics. The pool contains valid candidates, and the calibrated selector can recover valid outputs from the same pool at high budget. The failure is therefore not just bad sampling. It is the interaction between a proposal pool containing multiple candidate types and a proxy energy whose lowest tail is occupied by shortcut artifacts.

The third alternative is that the energy is useless. That is also too simple. The candidate-level Spearman correlation between negative energy and true score is positive over the full diagnostic set. The energy has information in the bulk. The failure appears because selection does not sample a typical candidate from the bulk; it chooses an extreme candidate from the lower tail. This is why a paper that reports only average correlation would miss the danger. A useful proxy can still be unsafe as a high-budget selector.

The fourth alternative is that diversity should solve the issue. The diversity-constrained selector tests this directly. It penalizes repeated modes, yet high-budget validity remains zero and artifact rate remains one. This means the artifact tail is not just one duplicated sequence. It contains enough distinct candidates that a diversity penalty can still select from the wrong class. This distinction matters for real systems, where adding diversity to beams or samples may improve surface variety without fixing the proxy tail.

The fifth alternative is that a stronger global penalty would make the result go away. The global-weight sweep weakens that objection. Increasing the global penalty raises selected energy and improves partial true score, but validity does not recover in the tested range. The diagnostic therefore supports a more careful conclusion: global constraints help, but the complete deployed energy must still be audited under candidate-pool pressure.

The sixth alternative is that the calibrated repair is unfairly tuned. The paper addresses this by reporting failures in the repair grid. Low-tail clipping without a shortcut penalty still fails. Stronger proxy penalties can recover validity in this diagnostic, but the manuscript does not claim that those settings are universal. A real model would need a predeclared proxy feature or a heldout calibration set. The grid is evidence that the failure is repairable when the shortcut is visible, not a claim that the proposed repair solves all EBT inference.

Appendix AP: Acceptance-Grade Real-EBT Extension

If this diagnostic were extended into a full real-EBT submission, the first upgrade would be trained checkpoints. The paper would train or obtain at least one Transformer energy model on a task where candidate-pool selection is a plausible inference method. The task should have hidden evaluation labels that are unavailable to the energy selector. The model should be evaluated at several candidate budgets, including a budget large enough to expose tail behavior. The same candidate pools should be reused for all selectors.

The second upgrade would be a stronger selector suite. Minimum energy would remain the stress target. Diversity selection would test whether surface variety helps. A calibrated energy selector would test whether tail clipping helps. A verifier-selected upper bound would show what is possible if hidden or external information is available. A random selector would identify the base quality of the generator. A learned repair could be included only if it does not train on the final hidden labels used for evaluation.

The third upgrade would be human-readable failures. For each task, the paper would show examples at low budget and high budget, including both selected failures and repaired successes. The example analysis should identify the feature that fooled the energy: repeated local pattern, premature termination, format regularity, short-range consistency, spurious attention concentration, or another domain-specific shortcut. Without examples, reviewers cannot tell whether the failure mode is meaningful or merely a metric artifact.

The fourth upgrade would be calibration by energy quantile. The real paper would bin candidates by energy and report hidden utility, artifact frequency, and proxy-feature activation in each bin. A dangerous model is one where the lowest-energy bins have worse hidden utility than moderate-energy bins. This quantile analysis would directly connect the synthetic order-statistic argument to empirical model behavior.

The fifth upgrade would be compute reporting. The paper should state candidate budget, number of prompts, number of seeds, hardware, runtime, and memory footprint. It should distinguish training cost from audit cost. A practical audit is valuable only if other researchers can run it. The present repository keeps the diagnostic CPU-light; a real-EBT extension should preserve that auditability even if model training itself is more expensive.

Those upgrades are not required for the narrow claim of this artifact, but they are the path from diagnostic submission to stronger empirical paper. The v4 manuscript now makes that path explicit and adds a small real-data tier so reviewers can see the boundary between what is proved here and what remains for a trained-model extension.

APPENDIX AQ: FINAL EVIDENCE SUMMARY

The final evidence chain is intentionally redundant. The proposition explains why a lower-energy shortcut component can dominate a selected minimum when the candidate budget grows. The base experiment shows that this happens in the controlled Transformer-structured landscape. The high-budget expansion shows that the collapse remains at $N = 512$. The shortcut-prior sweep shows that rare artifacts can still dominate final selection. The global-penalty sweep shows that strengthening one global term is not automatically enough. The repair grid shows that a proxy-only selector can escape the dangerous tail when it uses a visible shortcut feature. The candidate-level calibration analysis shows why bulk energy usefulness does not imply tail safety.

These pieces answer different reviewer questions. The proof answers whether the mechanism is coherent. The base curves answer whether the implementation exhibits the mechanism. The perturbation sweeps answer whether the result is a single fragile setting. The repair grid answers whether the tail diagnosis is actionable. The calibration plot answers whether the energy is merely useless or specifically unsafe in the selected tail. The failure-case rows answer whether aggregate averages hide the selected examples.

The submission should therefore be judged on a narrow but complete claim: minimum-energy candidate selection is not automatically reliable for Transformer EBMs just because lower energy looks desirable. If the low-energy tail is locally compatible but globally invalid, extra candidate budget can amplify the wrong candidates. The diagnostic does not end the real-EBT question, but it gives reviewers a concrete test that real systems should pass before deploying energy-minimizing candidate-pool inference.

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